Enabling LFRic on Isambard 3
A Reproducible AArch64 Software Stack Using Spack

Liam Berrisford[1], Krisztian Kohary[2], Denis Sergeev[3], Kolen Cheung[1], Nathan Mayne[2]
1 Research Software Engineering, University of Exeter
2 Department of Physics and Astronomy, University of Exeter
3 School of Physics, University of Bristol

Abstract

We present an automated workflow for building and validating the LFRic applications software stack on Isambard 3 using Spack environments. This work represents a first practical step towards making the LFRic stack portable across `*linux-aarch64*` platforms. The project integrates curated package repositories, enforces consistent compiler and MPI selections, and resolves key dependency chains. The workflow provisions a local Spack instance, generates environment manifests, performs staged installs to avoid known race conditions, and regenerates environment views for runtime use. We demonstrate repeatable deployment of LFRic Apps with Rose and CYLC tooling on Isambard 3 and provide diagnostics that simplify debugging and future site adaptation.

Workflow Step	Description
1. Concretize Spack Env	resolve the abstract environment into a fully concretised dependency graph;
2. Staged Install	stage and install packages in dependency order;
3. Regenerate View	regenerate the unified environment view to expose installed software; and
4. Validate	validate the build and runtime configuration to ensure correctness and reproducibility by running the `*lfric_atm*` canned example.

```
[+]  [    ]  lfric-apps-isambard@0.1.0 %none arch=linux-sles15-neoverse_v2
[+]  [bl  ]      ^mpich@4.3.2 %gcc@12.3.0/min757gnzig2eeh6ew4wdfzif2j6zneq arch=linux-sles15-neoverse_v2
[+]  [bl  ]      ^hdf5@1.14.6 %gcc@12.3.0/min757gnzig2eeh6ew4wdfzif2j6zneq arch=linux-sles15-neoverse_v2
[+]  [bl  ]      ^netcdf-c@4.9.3 %gcc@12.3.0/min757gnzig2eeh6ew4wdfzif2j6zneq arch=linux-sles15-neoverse_v2
[+]  [bl  ]      ^netcdf-fortran@4.6.2 %gcc@12.3.0/min757gnzig2eeh6ew4wdfzif2j6zneq arch=linux-sles15-neoverse_v2
[+]  [bl  ]      ^xios@2252 %gcc@12.3.0/min757gnzig2eeh6ew4wdfzif2j6zneq arch=linux-sles15-neoverse_v2
[+]  [bl  ]      ^metomi-rose@2.5.1 %none arch=linux-sles15-neoverse_v2
[+]  [bl  ]      ^cylc-flow@8.6.2 %none arch=linux-sles15-neoverse_v2
[+]  [bl  ]      ^cylc-rose@1.7.0 %none arch=linux-sles15-neoverse_v2
[+]  [bl  ]      ^cylc-uiserver@1.8.3 %none arch=linux-sles15-neoverse_v2
```

Legend
[+] installed in the environment
[bl] built from source (no binary)
^ dependency of the root spec
@ version
% compiler used to build
arch= target architecture

Figure 1: Built Environment
Example trimmed output of `*spack find*` showing the concretised LFRic build environment, including root specification, dependency tree, compiler toolchain, and target architecture for the installed stack.

Motivation

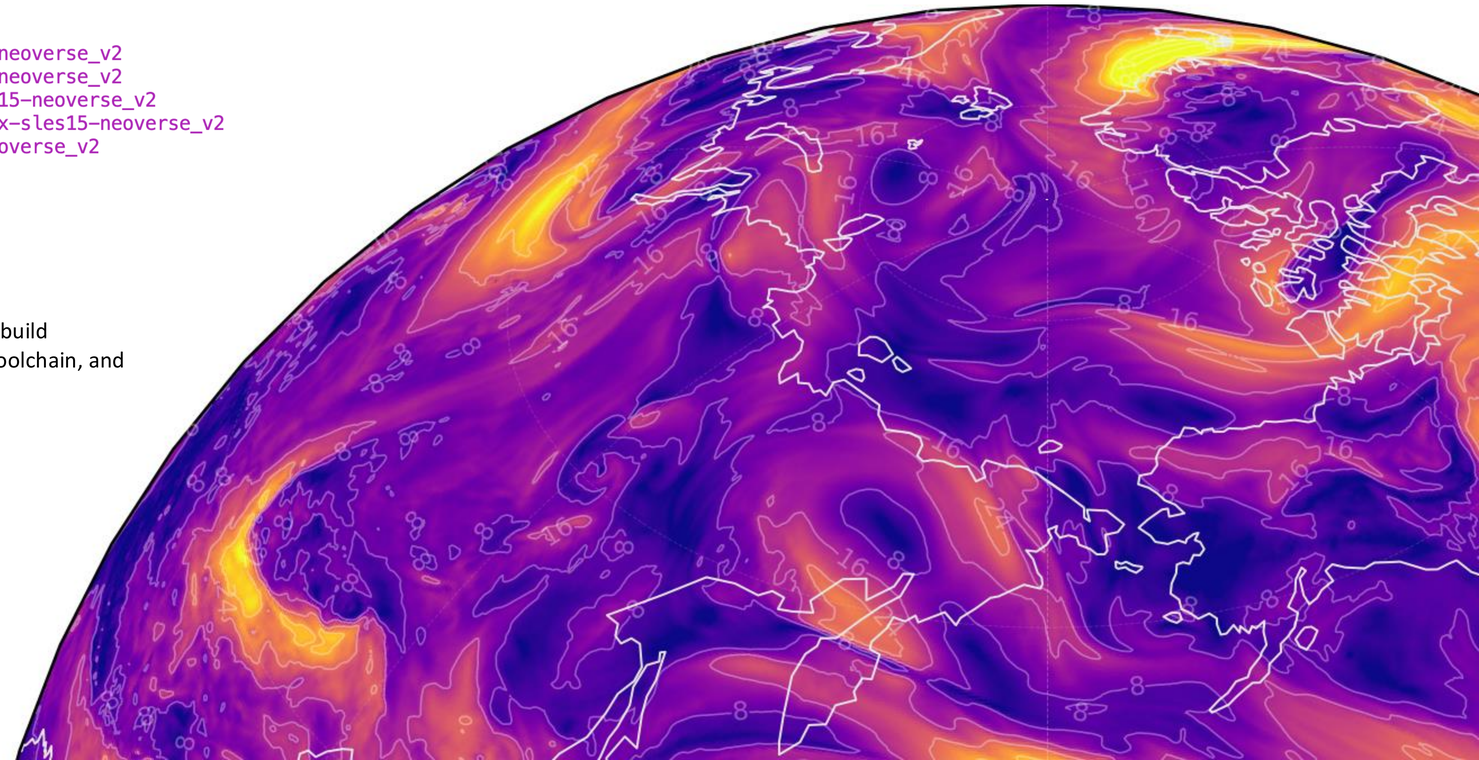
LFRic is a next-generation atmospheric model used for weather and climate prediction on Earth and other planets.
aarch64 portability across HPC sites needs reproducible, auditable builds.
Isambard 3 provides a realistic target for first-principles deployment, including operational constraints such as 24-hour job limits affecting CYLC workflows.

Results

Repeatable deployment of LFRic Apps on Isambard 3.
Consistent runtime environment via regenerated views.
Clearer pathway for portability to other linux-aarch64 sites.

Reproducibility and Verification

Concretised dependency graph fixing compiler/MPI at the root specification.
Deterministic environment regeneration with full provenance capture.
Short-run smoke test confirming toolchain integrity and runtime consistency.



Next Steps

(1) NVHPC-based environment: port the Spack env, validate compiler/MPI parity, and document deltas vs GCC.
(2) Expand suite coverage: add additional LFRic science suites and longer runs.